



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Conceiving services

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***Services* characteristics (*not just applications*)**

They work always: **resilient** and **secure**

They work independently of the number of users: **elastic**

Their huge architectural complexity is **transparent** to the user

They require a **business model**



From *technology delivery* towards *service delivery*

The alignment of information technology, business methods and organizational practices influence digital innovation, productivity and client satisfaction in service industries

Education models must support the transition from a technology delivery approach towards a service delivery approach based on business goals



First, WHY?

Then, HOW?



Why?



Motivations

- **Money** → *Better than the state of the art*
 - Novel
 - Higher quality
 - Lower cost
- **Ethical free service** → *Find free resources*
- ...

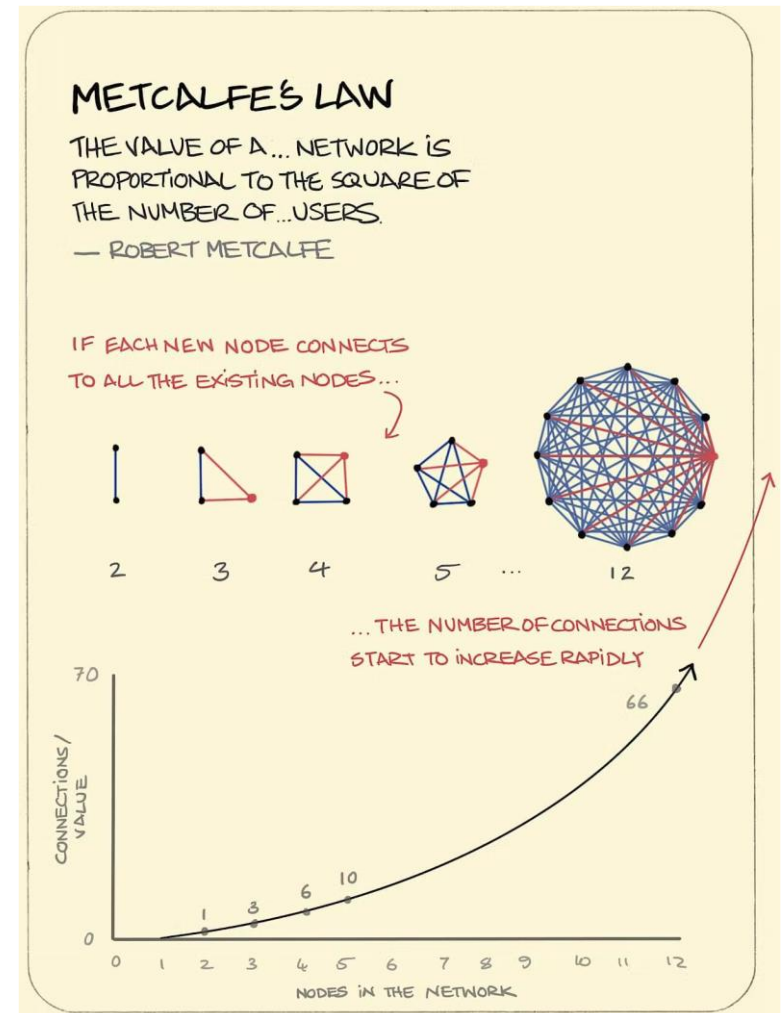
You have to know the main characteristics of digital services and their business models



The two main “laws” ruling the digital world

1) Resources: digitalization and dematerialization

2) Network: Metcalfe's Law



Economic consequences

DIGITAL



Metcalfe

- Data economy + Network economy (*Social economy*)
- Property(?) of immaterial resources
- Post-scarcity epoch (demonetization)
- Disintermediation provider-customer (through digital platforms)



Exponential growth and decrease



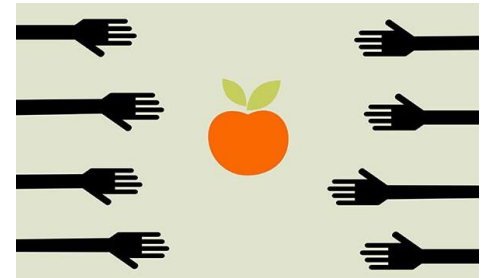
Post-scarcity model

The reasons for post-scarcity are **technological** ...

- storage costs continue to decline
- Internet is ubiquitous
- communications continue to be more affordable for all

... but the consequences are **business-related**

A fundamental **human need and competition** exists in a world of **limited resources** and costs can be factored accordingly (sometimes, they have to invent scarcity even when there is none)



Demonetization → *post scarcity*

From the scarcity of physical resources and costs of production and logistics to the unlimited reproducibility and (practically) no cost of distribution of digital resources

- Music
- Movie
- Email
- Phone calls
- Online shops
- Information
- Free software
- Photo
- ...



Which business model?



Post-scarcity model: Kodak and pre-digital photography

- In **1996**, Kodak had sales of \$15 billion
- In **2002**, it was bankrupt (with 120,000 layoffs)
- Was it possible that no one had noticed the arrival of digital?
The explanation is different → *storage more than digital*

Business consequences on the entire supply chain:

- Every time a photo was taken, there was an incremental cost associated with it (film, processing, delivery/withdrawal) → the more pictures that were taken, the higher the price
- The digital photography is a post-scarcity model in which goods can be created in volume with minimal human effort and minimum cost
- Digital photos uploaded from a mobile phone to Instagram have an almost identical cost if we upload a single photo or an entire album



Disruptive consequences

Disruption relates to the creation of a new market and the displacement of existing players

There are examples of disruption in all industry verticals where the status quo is rewritten, and customers reset their expectations

Blockbuster → Lovefilm → Netflix

- **Blockbuster** provided home video and video game rental services
- **LoveFilm** was a provider of DVD-by-mail services; then started a video-on-demand service (acquired by Amazon in 2011) → DVD by post was an evolution but not a disruption; VoD service was too late
- **Netflix** created an entire new **disruptive** market by introducing streaming media and video on demand → leveraging top technologies of that time

Then, in 2013, Netflix began to produce original content, beginning with *House of Cards*. By 2022, original productions accounted for half of Netflix's library in the United States



Not only ideas, but also the ability to execute

For example, Netflix has always been obsessed by the quality and resiliency of its network-based services

- Architectures with **no system bottleneck**
- Architectures with **no single point of failure**
- **Extreme resilience management**
Netflix created the famous **Chaos monkey** software that arbitrarily kills server resources across production infrastructure, thus seasoning operational staff to the real challenges of what to do when somethings go wrong



Another example of demonetization

- The costs associated with making a call were dependent on the factors:
 - The distance between the caller and the receiver
 - The duration of the call
- As telephony technology evolved, a similar model existed with mobile phones: national calls and messages were cheaper than those done internationally
- Then VoIP arrived. And Skype offered free calls over the Internet between users irrespective of their geographical locations. Now, Microsoft, Google, Facebook offer similar services and all mobile operators had to alter their pricing structures
- Demonetization drives new societal norms: consumers can decide what they are willing to accept, based on different options available



Demonetizations and business consequences on competitors

- Similar demonetization exist for **Operating systems** (Windows vs Linux and Android) → MS had to alter its old approach: for the first time in the history, MS offered free upgrade from Windows 7 to Windows 10, and from Windows 10 to Windows 11
- After **Tesla**, several car manufacturers are now offering free software updates to their automobiles “over the air”, sometimes even for their navigation systems:
 - a huge change with respect to the expensive card or DVD purchases
 - an easy way with respect to the need of visiting a garage or a dealer

The entire process has been dematerialized, streamlined, automated and consequently demonetized through novel services, networking, security and software delivery. No competitor can survive using the “old material process”



Really disruptive (*not just evolutionary*) companies

- **Google:** let us organize the world's information
- **Tesla:** let us accelerate the transition to sustainable transportation
- **Facebook:** let us connect all the humans
- **Apple smartphone:** all digital services in your hand
- **SpaceX:** humans should become a multi-planetary species
- **Public cloud industry (AWS, Azure, GCP):** let us replace the local data centers
- ...



Dematerialization everywhere

The tendency is towards the **removal of most physical components**:

- In the case of Kodak, the digital camera removed the need to purchase camera film and roll development. The smartphone removed the camera
- With the emergence of iTunes and Amazon music, and Spotify, people did not need to buy CDs/DVDs
- Cloud computing and ubiquitous Internet have dematerialized the on-premise data centers, but also the home cinema and other markets: software and games are provided through the Internet and not through DVDs at a store
- Payment solutions have experienced two huge disruptive changes:
 - 1) Credit and debit cards avoided physical cash
 - 2) Apple Pay, Android Pay and Satispay avoid physical cards (try to explain the concept of a bank check to anyone under the age of 15: they will most likely think you are joking)



Another disruptive business: companies that leverage physical resources but do not own them

- **Uber**
- **AirBnB**
- **Deliveroo, UberEats, etc.**
- **Ted talks:** ideas worth spreading

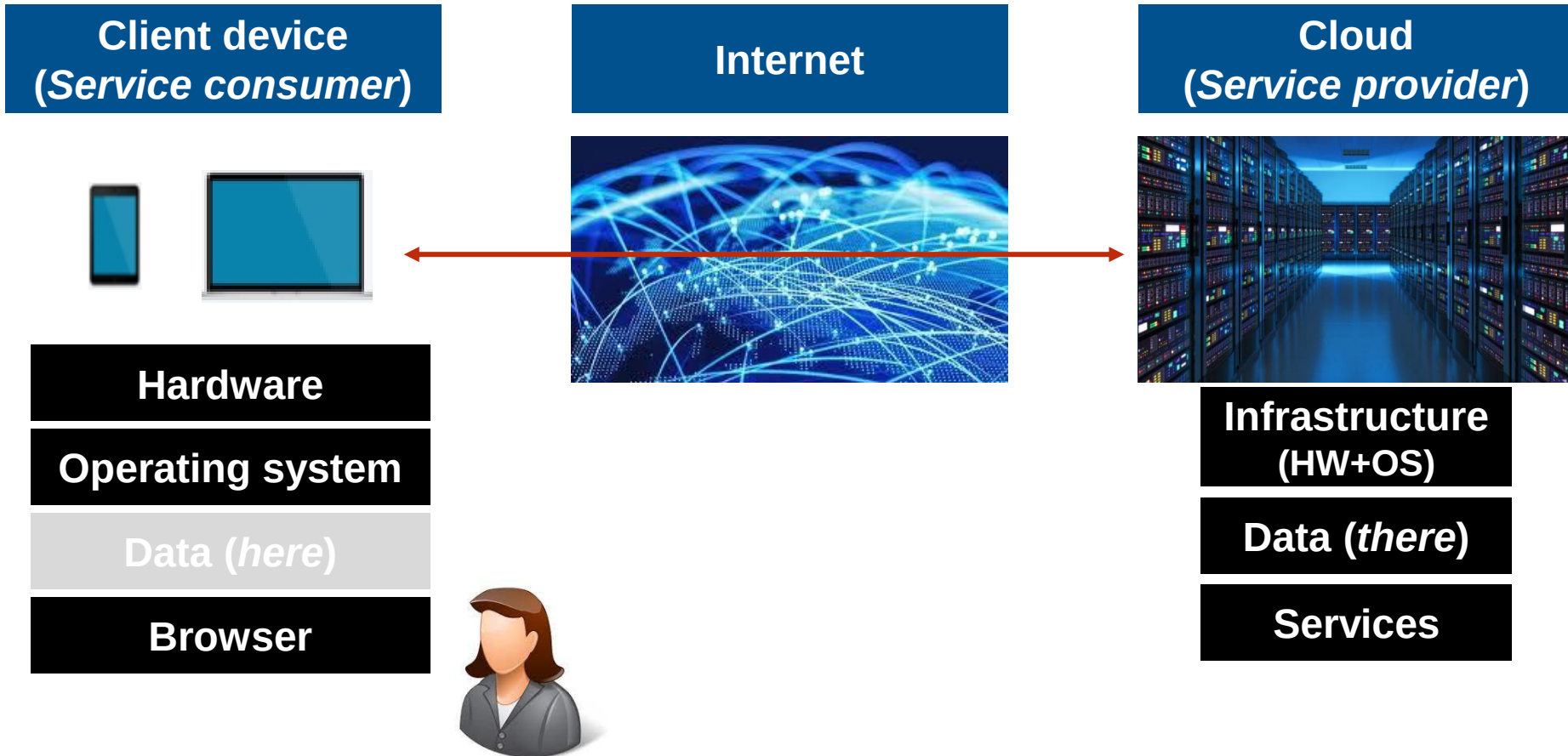
- But the top example of leveraging physical resources not owning them is ...



Phone, Camera, Flashlight, Voice and video recorder

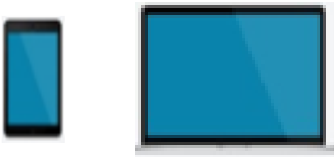
Game console, Alarm clock, Map, Health monitor, Millions of apps, ...

What I really own in the digital society?



What I really own in the digital society?

Client device
(Service consumer)



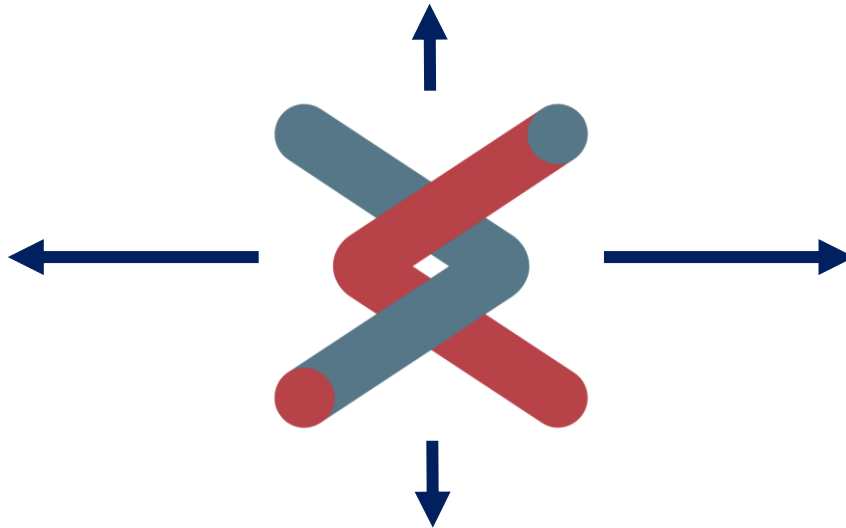
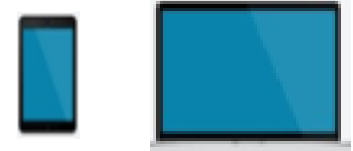
Hardware

Data (here)



Property and digital society

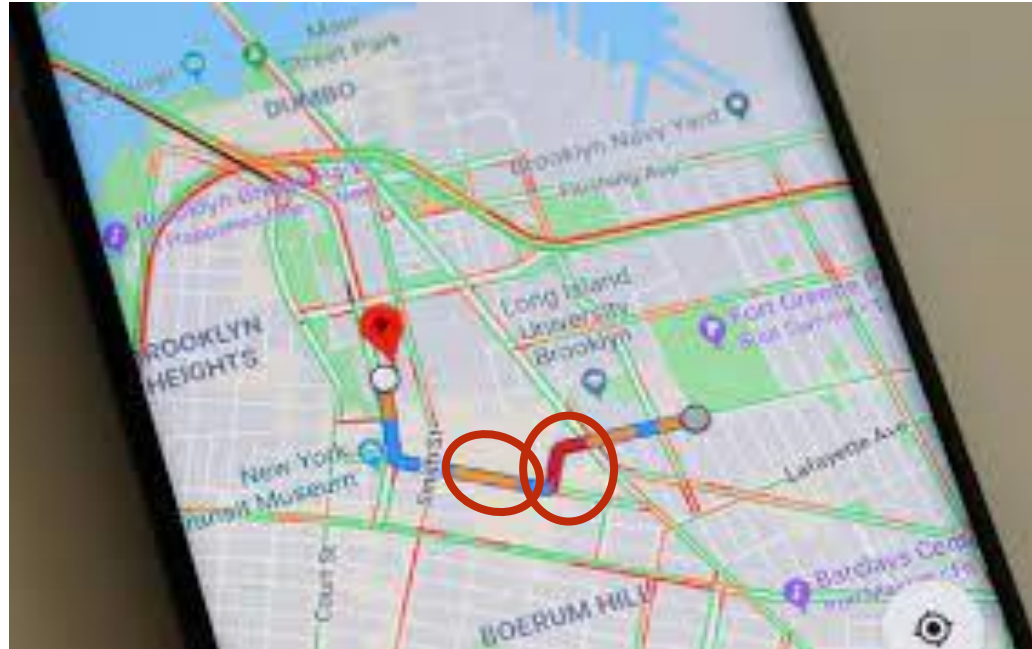
- A smartphone, a laptop should not be considered as an **absolute property**
- Each of them is a **nexus** that is, an important connection between the parts of a system or of a group of things, persons, or events that are parts of a chain of causation (e.g., King's Cross is the nexus of the London subway)



Metcalfe's Law: the value of a network is proportional to the square of the number of connected users of the system



An example of *social* value



Metcalf's Law: the value of a network is proportional to the square of the number of connected users of the system (n^2)

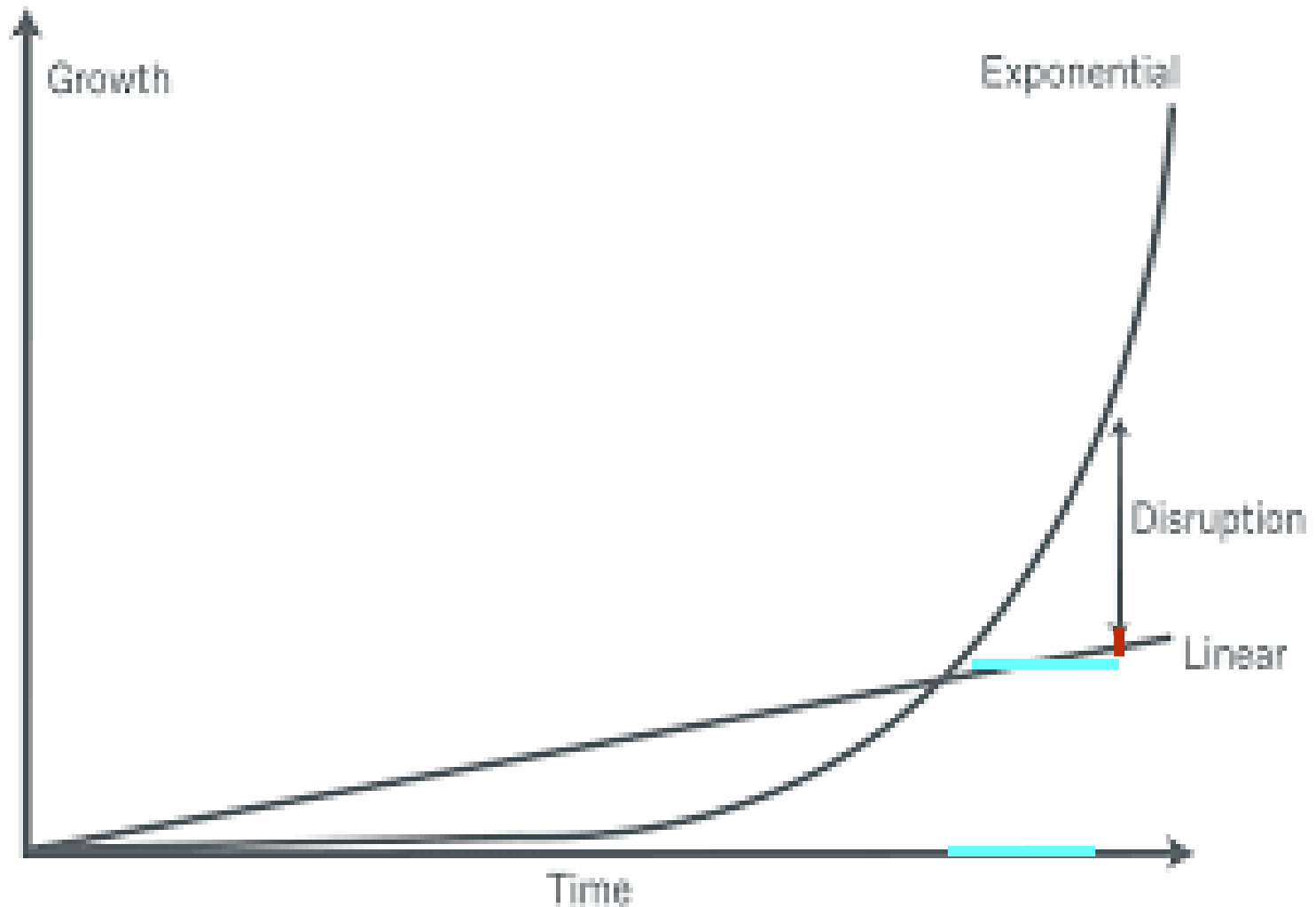


Democratization

- Costs drop and availability increases to a level where services and goods are available to everybody
- Democratization is the logical next step where **demonetization** and **dematerialization** have taken place
- Democratization brings services to the “masses”:
 - 1) **Users**
 - 2) But also **startups** (thanks to cloud) can leverage the “as a service”-based solutions with just one good Internet connection and a credit card. Many novel companies are offering service capabilities that exist 100% in the public cloud



Living in exponential times



In exponential times, the first wins all!

(An old motivation for poor software. Today?)

Being first is the crucial factor for success. With first-mover advantage, you can leverage data engineering, and exponential factors of network effects and related marketing

- In the interconnected era, the most innovative companies will support a compelling vision for a product or service by using higher order components built on the power of existing platforms
- Developing a wide understanding and vision of what is possible and then crystallizing that vision into technology (that is, implementing it) will be the key skill for companies in this era
- The winners will be companies that win the race toward finding the highest-level componentry of technologies and services to support a disruptive business model



1 – Computing era

COMPUTING ERA

The mass market computing era took off in the '80s, sparking unprecedented productivity gains. IBM made mainframes a must-have for corporations, and "Wintel" democratized computing. But when it came to connectivity, there were still significant limits. Prior to the computing era of the 80s, the automated capturing, processing and distributing of data only happened on a proprietary computer-by-computer basis. As a result, information still needed to be manually moved from system-to-system via tape or disk.

1982

Time Magazine declares computer the "machine of the year"



Complex proprietary networks like SNA or X.25, which provide emulation or limited data transfer options, create "islands of networks"



1984

Apple launches Macintosh and the term "cyberspace" is coined



The tongue-in-cheek term "sneakernet" acknowledges that connectivity still basically amounts to someone wearing sneakers carrying data from one computer to another



2 – Networked era

NETWORKED ERA

Businesses needed a new breed of connectivity to succeed. Data networks grew with the mass market adoption of PCs, transforming computers from standalone boxes to connected systems. File servers and local area networks (LANs) made it faster and easier to share information. Telecommunications companies laid new network lines that circled the globe many times over. As more people, companies and information became connected via networks, the groundwork was laid for the biggest transformation of all: the Internet.

1993
Mosaic web browser released



1995
First item sold on eBay



1990

The **World Wide Web** is born when Tim Berners-Lee, a researcher at CERN, the high-energy physics laboratory in Geneva, develops HyperText Markup Language (HTML)



1994

Netscape and Yahoo founded



3Com, Cisco, Ciena and Juniper pioneer new switching and routing technologies



3 – Connected era

CONNECTED ERA

In the connected era, networks at last evolved into open thoroughfares for transmitting information around the world. People could access data beyond their own PC, LAN or WAN, quickly shrinking the distance between businesses. Founded in 1998 as a vendor-neutral environment where providers could come together and “peer” to exchange information, Equinix helped scale the Internet by enabling companies around the world to connect and collaborate. The networked era ultimately morphed into a time of constant “anytime, anywhere” connectivity, made possible by new mobile, cloud and social media technologies.

1997
Amazon
goes public



88
MILLION

According to the U.S. Department of Labor, in 1995 there were only **22 million Internet users** in the United States. By 1998, the figure had quadrupled to **88 million**.

1998
Equinix founded
to help scale the
Internet



2002
Friendster
Introduced as first
social network



2003
LinkedIn
formally
launches



2004
Facebook
founded

2005
Reddit
founded



2006
Twitter
launches

2010
Instagram
launches



4 – Interconnected era

INTERCONNECTED ERA

A new era is dawning in our digital economy: the interconnected era. In a world where cloud dominates and enterprise business models are interdependent, the demands placed on connectivity have reached a new high. That's because companies now forge advantage by collaborating in communities with other enterprises via secure, reliable and internetworked connections among many participants. To compete and succeed – today's businesses need interconnection, a form of connectivity that's instant, direct, massively scalable and increasingly more flexible, secure and reliable than its predecessors.



Mobile applications empower people to interact and work from anywhere. More emails are now opened on mobile devices than on computers

Global e-commerce revenues grow from \$286 billion in 2000 to nearly **\$1.5 trillion** in 2014

\$1.5 TRILLION

Cisco launches InterCloud



Gartner predicts 4.9 billion connected "things" in use by the end of 2015, growing to **26 billion** by 2020.



Intel unveils platform for Internet of Things



\$7.1 TRILLION

IDC estimates the Internet of Things sector will hit **\$7.1 trillion** by 2020



Equinix makes Microsoft Azure ExpressRoute available through data centers in **16 markets** globally

New connectivity-driven capabilities such as GPS-guided maps and near-field communication have spawned indispensable modern conveniences such as **Uber** and **Apple Pay**



From marquee enterprise customers to technology innovators like **Amazon** and **Microsoft**, thousands of enterprises and service providers worldwide make their way to Equinix data centers to interconnect



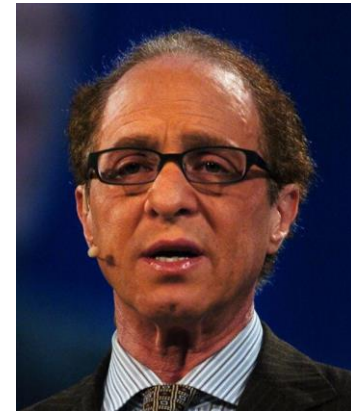
What about the 5th era?

Imagination, design, and marketing will become paramount. Computer science and engineering will take a **backseat** to data science and data engineering as first creators of differentiating value



The Kurzweil's thesis: “**The Law of Accelerating Returns**” (2001): “in the 21st century, we won't experience 100 years of progress; it will be more like 20,000 years of progress”

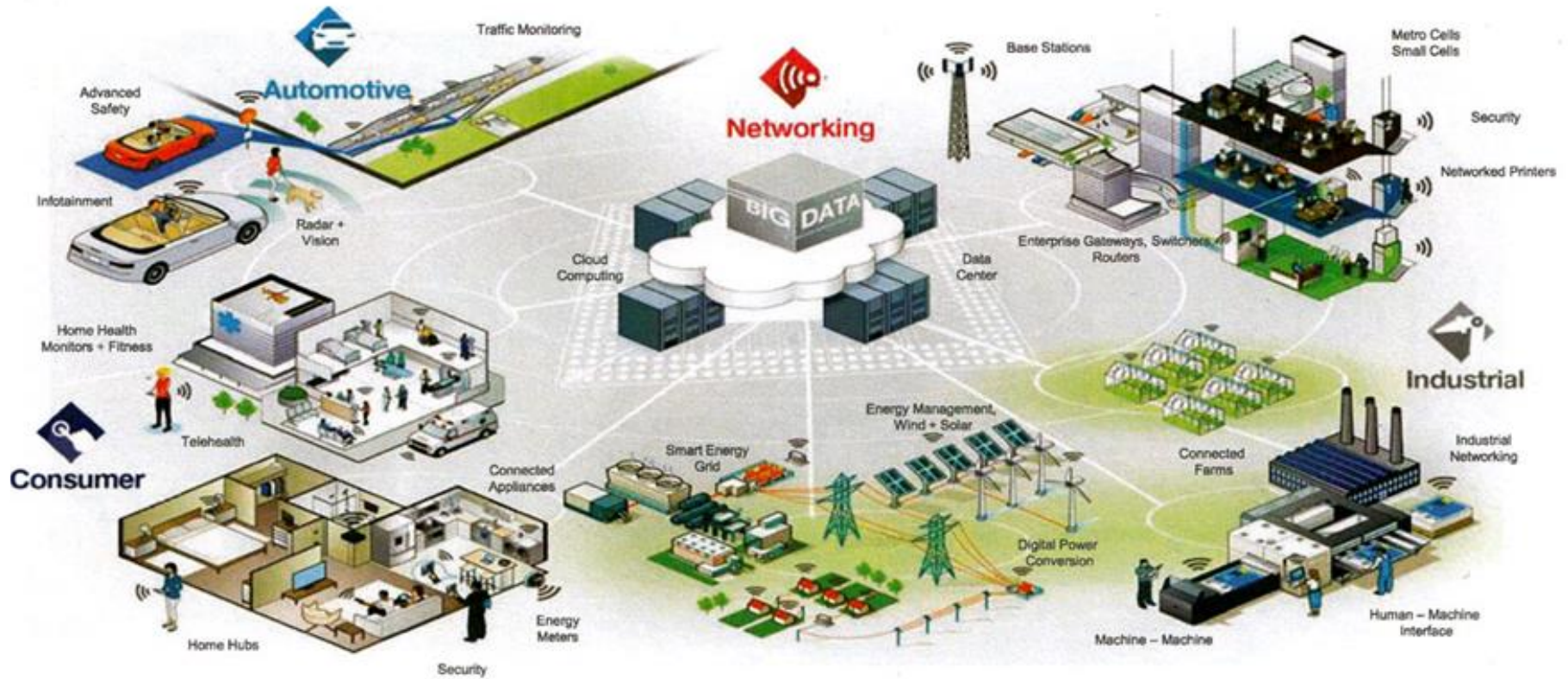
- At the beginning, the progress is made on basic components. This period is represented by the **flat part of an exponential curve**
- At some point, it becomes possible to **combine** components into higher level, more powerful constructs. At that point, progress or change accelerates, and these components can be combined into even more powerful components



Ray Kurzweil



Challenges and opportunities for the Cyber Physical world



We are connecting people, data, things, machines, ..., that are enhanced by a lot of **sensors** and **actuators**

We are increasing **self-automation** of smart objects because ...

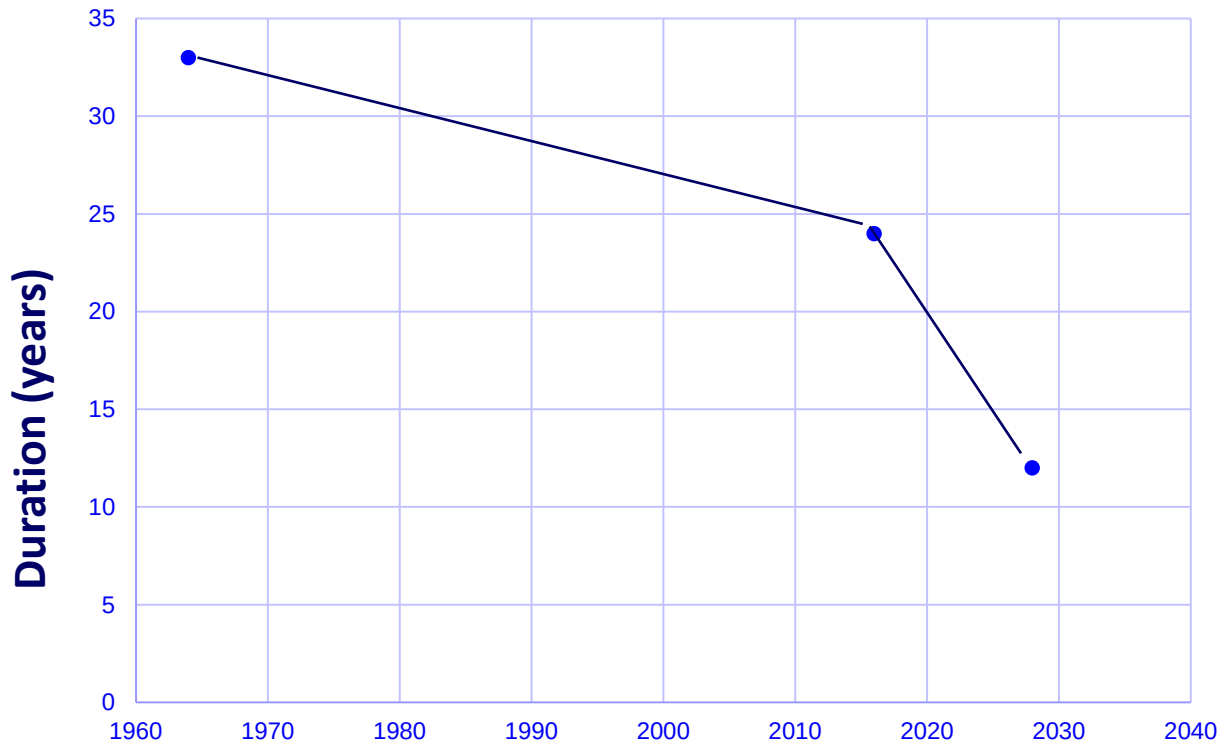


Exponential times

- There is no just **exponential growth** but also **exponential losses** in digital technology focused companies
- Wherever we look, most market-dominant digital firms in this era are *metaphorical children*
- **Rapid growth and rapid death are common in this era** although in many cases the conclusions are due to acquisitions → **This is the real goal of many startups!**



Average duration of companies [S&P data]



Fewer than 12% of companies included in Fortune's 1955 list were on the 2020 list

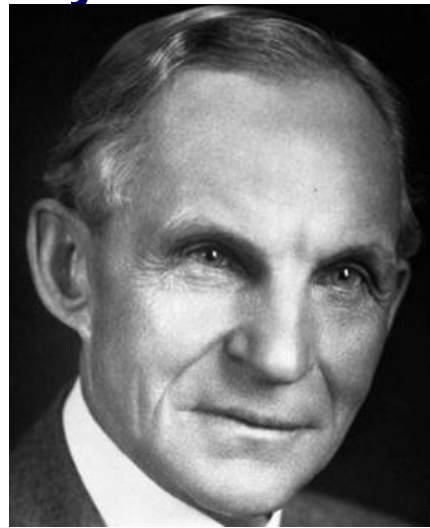
440 of 500 companies in 1955 went bankrupt or were acquired or merged by other companies

Who will (r)exist in 2050?



Innovation does not satisfy existing clients

- **The problem might lie with a maniacal focus on existing customer satisfaction** → companies are so fixated on ensuring that their existing clients are happy today that they are blind to evolving customer requirements and to the evolving emerging technologies that could facilitate change (Bower and Christiansen)
- There are many other reasons: bureaucracy, arrogance, tired executive, poor planning and short-term investment horizons, ...
- But often the fundamental reason is a paradox: leading companies succumb to the most popular management dogma: **“to have success, stay close to your customers”**



If I had asked people what they wanted, they would have said faster horses.

— Henry Ford —



The “technical debt” problem of some *oil tanker vessel* companies

The banking industry has always been considered a late adopter of promising technologies

- It is like an enormous **oil tanker vessel**: monolithic and unable to change direction at any pace
- Although there are some progressive financial branches, the common perception is that their pace of change is glacial: most is due to the legacy complexity, past technological investments, and burdensome regulation



The “no technical debt” opportunity of some *speedboat* companies

- The greenfield organizations in any vertical sector (e.g., retail, media, fashion) stand for diametrically opposite to banking stands
- If you can start from a “clean state” with zero “technical debt”, you can avoid most concerns regarding interoperability and application compatibility. So, they can create technology ecosystems that are unrecognizable to the companies with an established presence
 - They have no local infrastructures because they use public cloud services
 - They have no corporate devices because each employee uses his own equipment

These companies are the metaphorical *speedboats*: agile, fast, and able to manoeuvre quickly



Example: *Cloud computing*

The service and deployment models associated with cloud computing and virtualization have allowed niche software houses to provide carrier-grade infrastructure, network and security services on a per-usage, service-oriented basis

- This levelling of the playing field also brings tangible benefits to the customers that now are free to engage with large offering of high-quality business services
- It is difficult to adopt cloud-first culture coming from legacy models and businesses

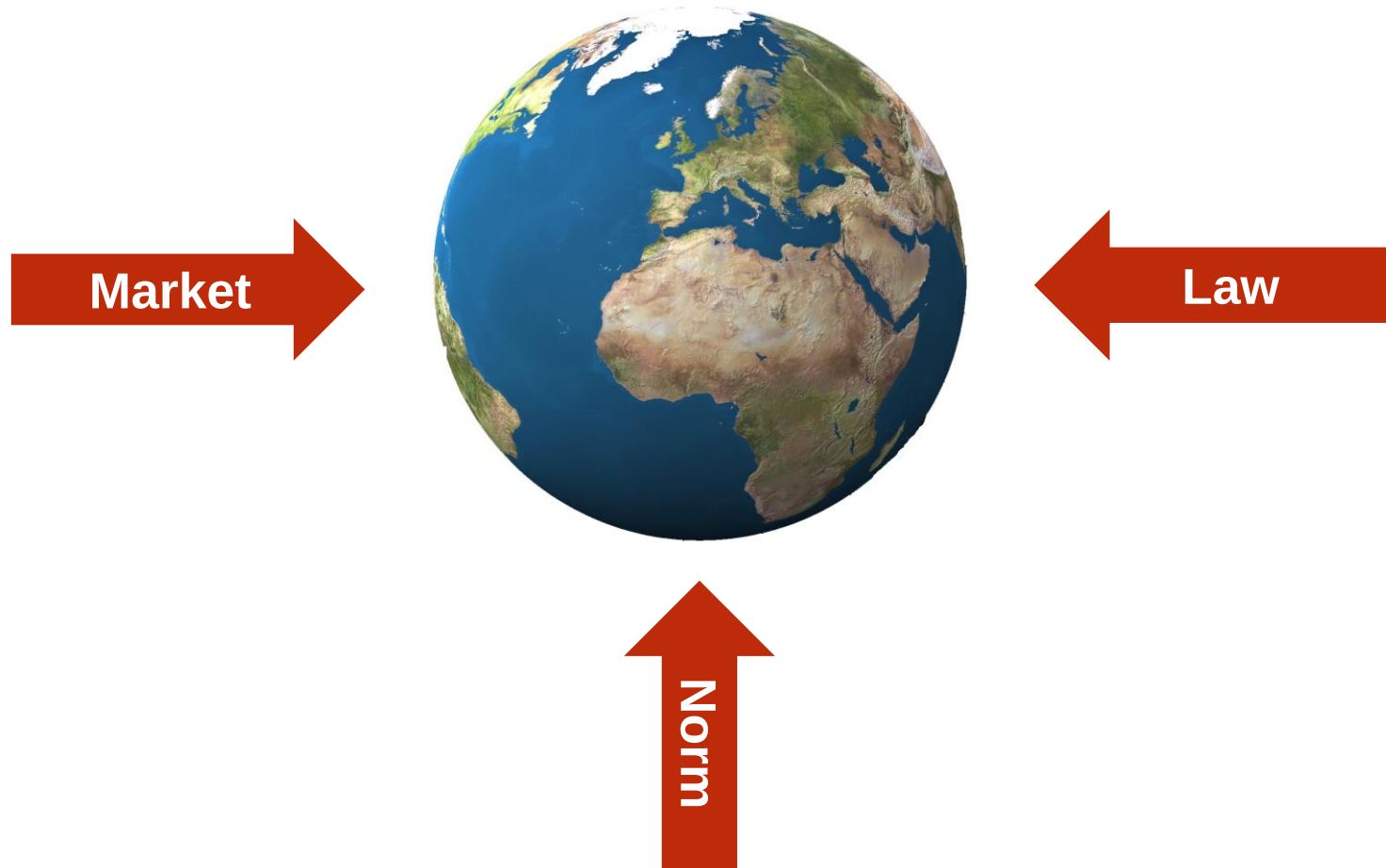


The risk of *iterative disruption*

- A salient although often overlooked point regarding disruption is that this process is **iterative**
- We are now seeing **disruptors** that risk of being **disrupted**, e.g.,
 - **Netflix** challenged by **Amazon** and **Disney**
 - **Uber** challenged in some countries by **Didi** (Chinese company with Alibaba and Apple behind) not only car owner drivers and delivery partners, but also collaborations with policymakers, taxi industry, automobile industry and local communities to solve the world's transportation, environmental and employment challenges through AI technology and localized smart transportation innovations
 - **Google search** challenged by **ChatGPT** (belonging in large part to Microsoft) → forced to anticipate the exit of **Google Bard** (*some unfortunate consequences*)



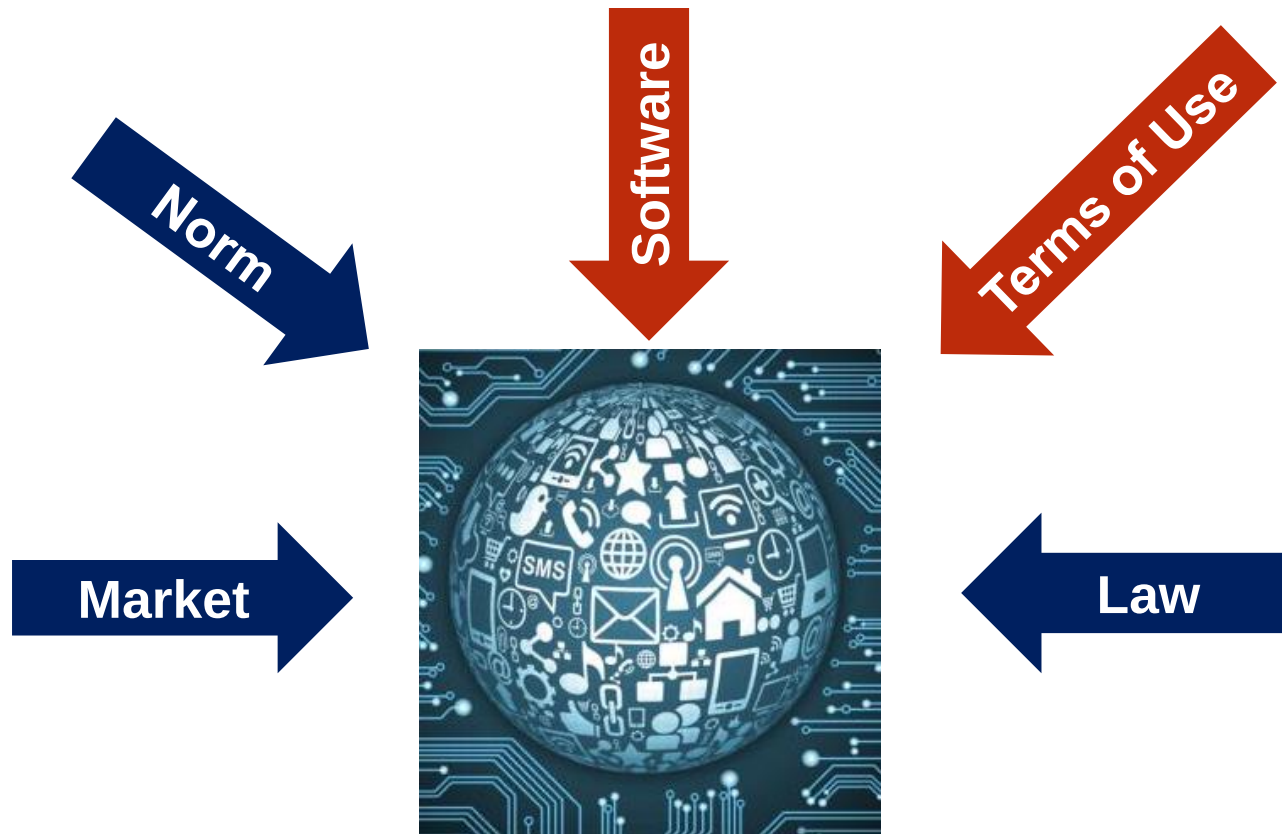
Forces governing the *physical* society



In the *physical society*, it is unlikely or impossible that the governing forces may be in opposition for long. **At the end, the Law prevails**



Forces governing the *digital* society



In the *digital society*, forces might remain in opposition for long. It is necessary to evaluate from time-to-time which force is (really) determining the rules in that period

Example: the power of *Terms of Use*

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Search



Statement of Rights and Responsibilities

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1. For content that is covered by intellectual property rights, like photos and videos (IP content), you specifically give us the following permission, subject to your [privacy](#) and [application settings](#): you grant us a non-exclusive, transferable, sub-licensable, royalty-free, worldwide license to use any IP content that you post on or in connection with Facebook (IP License). This IP License ends when you delete your IP content or your account unless your content has been shared with others, and they have not deleted it.



Summary

The novel technology paradigms (digital + Internet) have changed the way modern enterprises (should/could) conduct business, but exponential times, dematerialization, demonization represent tough challenges for any company, old and new

The digital-Internet society rule is:

“If you do not *disrupt* periodically your business, someone else will”

(No one wants to be the next Kodak!)



Possible business models

- **Free**
 - **Paid by personal data**
 - **Paid by advertising**
- **Registration and All you can eat**
- **Pay per use**
- **Sharing business (“You produce, I sell and get 30%”)**
- **Gig economy (platform+on-demand autonomous workers)**
- **...**
- ***Others yet to come***



Customer consequences (*if the service wins*)

DIGITAL



Metcalfe

- **Service more important than property**
- **“Free” services**
- **Social visibility vs Privacy**
- **Limited ownership of digital devices → *Term of Service / Term of Use***
- **Computer is not relevant. So, it is often masked**
 - the information on computers is important
 - the service provided by computers is important, however, a successful service must work always
- **Ease of use, but complete opacity on algorithms and data management**



Transparency has a countereffect

- Masking the complexity of the technology and foundational algorithms simplify the digital life
- The consequence is that consumers have little awareness and no insight with which to make an informed market decision
- **Consumers should be given all the awareness and instruments with which to make informed decisions**
- The situation is analogous to the forces that resulted in the establishment of nutritional labeling for foods
“Nutritional labeling” about digital risks or a cyber version of self-certification will help focus the attention of all parties on its critical importance for a person or a company
- Placing the cybersecurity and reliability burden on the user is an unrealistic expectation, yet it is a major tenet of present activities



Information-enabled enterprises

- As an industry becomes information-enabled and powered by information flows, its price/performance begins doubling approximately annually or every two years
- The technology stack used to deliver successful digital solutions is known as **SMAC**:
 - **Social**
 - **Mobile**
 - **Analytics**
 - **Cloud**
- Each of these components has a profound impact on the methods that companies can use to reach their customers



Manipulative and seductive power of technology



**TIME: the only truly
non-renewable resource!**



Stay with me! is the modern business. Still valid in the future?

"Oh, won't you stay with me?

'Cause you're all I need

This ain't love, it's clear to see

But darling, stay with me"

